



HSP

Historical Service Performance (HSP) is a platform containing historical data from Darwin, with an API to allow easy querying of service performance data.

You can query for all trains between a pair of locations, between two times and on two dates and filter the results by train operator.

Accessing the API

To access the HSP API, you must have an account on the National Rail Data Portal at <https://opendata.nationalrail.co.uk/>. If you already consume real-time data from Darwin, you will already have an account.

After logging in and/or signing up, ensure the 'HSP' box is checked under 'Subscription type'.

Using the API

There are two API calls available - 'serviceMetrics' and 'serviceDetails', and documentation is available at <https://hsp-prod.rockshore.net/>.

For each API call, you must set the HTTP header Content-Type: application/json, and provide an HTTP Authorization header with the username and password for the National Rail Data Portal.

The HSP API does not support CORS and doesn't return an Access-Control-Allow-Origin header. This means you can't use it with JavaScript from a browser on another domain.

Service Metrics

The serviceMetrics call is used to request a summary of service performance for a particular journey. It requires an HTTP POST call with a JSON hash containing the following keys:

- *from_loc* and *to_loc* are valid three-letter codes (CRS codes) for the origin and destination locations on a journey
- *from_time* and *to_time* are the times of the day you wish to query and must be in the format HHMM, e.g. 0700
- *from_date* and *to_date* are the historical dates on which you wish to query and must be in the format YYYY-MM-DD
- *days* must be either WEEKDAY, SATURDAY or SUNDAY, and will limit the query to trains which run Monday to Friday, or on Saturday, or on Sunday

- *toc_filter* (optional) must be a list of TOC Codes
- *tolerance* (optional) is a set of up to three values to set the tolerances, e.g. ['2', '5', '10'] will return statistics for the number of trains which arrived at the *to_loc* within 2 minutes, within 5 minutes and within 10 minutes of their scheduled time

```
POST https://hsp-prod.rockshore.net/api/v1/serviceMetrics HTTP/1.1
Authorization: Basic {Base64("email:password")}
Content-Type: application/json
Host: hsp-prod.rockshore.net
Content-Length: 178

{
  "from_loc": "BTN",
  "to_loc": "VIC",
  "from_time": "0700",
  "to_time": "0800",
  "from_date": "2016-07-01",
  "to_date": "2016-08-01",
  "days": "WEEKDAY"
}
```

The response will be a JSON object containing:

- A *header* with the *from_loc* and *to_loc* of the query
- A *Services* containing a *serviceAttributesMetrics* with the *origin_location* and *destination_location* of the train, the *gbtt_pta* and *gbtt_ptd* (public arrival and departure times), the *toc_code* of the operator of the service, and a *matched_services* containing a list of RIDs of all trains. These RIDs can be used in the *serviceDetails* call to get further information.
- A *Services* containing a *Metrics*, with the *tolerance_value* selected, a *num_not_tolerance* (number of trains outside this tolerance), a *num_tolerance* (number of trains within this tolerance), and a *percent_tolerance* and *global_tolerance*

```
{
  "header": {
    "from_location": "BTN",
    "to_location": "VIC"
  },
  "Services": [
    {
      "serviceAttributesMetrics": {
        "origin_location": "BTN",
        "destination_location": "VIC",
        "gbtt_ptd": "0712",
        "gbtt_pta": "0823",
        "toc_code": "GX",
        "matched_services": "22",
        "rids": [ "201607013361753", "201607043443704", ... ]
      },
      "Metrics": [
        {
          "tolerance_value": "0",
          "num_not_tolerance": "5",
          "num_tolerance": "17",
          "percent_tolerance": "77",
          "global_tolerance": true
        }
      ]
    },
    {
      "serviceAttributesMetrics": {
        "origin_location": "BTN",
        "destination_location": "VIC",
        "gbtt_ptd": "0729",
        "gbtt_pta": "0839",
        "toc_code": "GX",

```

```

        "matched_services": "22",
        "rids": [ "201607013361763", "201607043443714", ... ]
    },
    "Metrics": [
        {
            "tolerance_value": "0",
            "num_not_tolerance": "7",
            "num_tolerance": "15",
            "percent_tolerance": "68",
            "global_tolerance": true
        }
    ]
},
...
]
}

```

Service Details

The serviceDetails call is used to request details of a particular train by using its RID. It requires an HTTP POST call with a JSON hash containing a *rid* with the RID of the service in question.

```

POST https://hsp-prod.rockshore.net/api/v1/serviceDetails HTTP/1.1
Authorization: Basic {Base64("email:password")}
Content-Type: application/json
Host: hsp-prod.rockshore.net
Content-Length: 33

{
  "rid": "201607294212242"
}

```

The response will be a JSON object containing:

- A *serviceAttributesDetails* containing the *date_of_service*, *toc_code* and *rid* of the train, along with a list of *locations* containing the *location* code, *gbtt_pta* and *gbtt_ptd*, the *actual_ta* and *actual_td* (*actual time of arrival and departure*), and if the train was delayed or cancelled, a *late_canc_reason* with the reason for the delay or cancellation

```

{
  "serviceAttributesDetails": {
    "date_of_service": "2016-07-29",
    "toc_code": "GX",
    "rid": "201607294212242",
    "locations": [
      {
        "location": "BTN",
        "gbtt_ptd": "0712",
        "gbtt_pta": "",
        "actual_td": "0711",
        "actual_ta": "",
        "late_canc_reason": ""
      },
      ...
      {
        "location": "GTW",
        "gbtt_ptd": "0749",
        "gbtt_pta": "0747",
        "actual_td": "0751",
        "actual_ta": "0744",
        "late_canc_reason": ""
      },
      {
        "location": "VIC",
        "gbtt_ptd": "",
        "gbtt_pta": "0823",

```

```
    "actual_td": "",
    "actual_ta": "0823",
    "late_canc_reason": ""
  }
}
```

Rules for Inclusion of Trains

For a given API request, HSP will only return data that matches all of the following

- The Public Departure time at boarding location is identical to the service
- The Public Arrival time at alighting location is identical to the service
- TOC code is identical to the service
- Service not marked as deleted
- Date ran is within specified date range
- Day of week ran matches specified day type
- An actual arrival time has been recorded at the alighting location (if the location hasn't been cancelled)

Help!

For help in using the API, please post to the [openraildata-talk](https://groups.google.com/forum/#!forum/openraildata-talk) (<https://groups.google.com/forum/#!forum/openraildata-talk>) mailing list, where other users of the service can help you.

National Rail Enquiries Data Feeds	
Data Feeds	About the Feeds • Darwin Webservice (Public) • Darwin Webservice (Staff) • Historical Service Performance • Push Port • KnowledgeBase • DTD • Locations (PoC) • Real Time Journey Planner
LDB API	About
LDB-SV API	About
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DTD	About • Fares • Timetable
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